The Chain Rule

MATH 1334 October 19, 2026

Differentiating composite functions

Question. How do we differentiate a composite function?

$$F(x) = (f \circ g)(x) = f(g(x)).$$

Identify the outer function f and the inner function g. Here u stands for the input to the outer function.

1. $F(x) = \sin(2x)$.

$$f(u) = \sin u, \qquad g(x) = 2x, \qquad f(g(x)) = \sin(2x).$$

2. $F(x) = \sqrt{1+x^2}$.

$$f(u) = \sqrt{u}, \qquad g(x) = 1 + x^2, \qquad f(g(x)) = \sqrt{1+x^2}.$$

3. $F(x) = \sin\sqrt{x}$.

$$f(u) = \sin u, \qquad g(x) = \sqrt{x}, \qquad f(g(x)) = \sin(\sqrt{x}).$$

4. $F(x) = \cos^4 x$.

$$f(u) = u^4, \qquad g(x) = \cos x, \qquad f(g(x)) = (\cos x)^4.$$

In particular, $\cos^4 x$ means $(\cos x)^4$.

The chain rule

Theorem (Chain rule)**.** *Suppose g is differentiable at a and f is differentiable at $g(a)$. Then $F = f \circ g$ is differentiable at a, and*

$$F'(a) = f'(g(a))\, g'(a).$$

Equivalently,

$$\frac{d}{dx} f(g(x)) = f'(g(x))\, g'(x).$$

Idea: Differentiate the outer function, keeping its input unchanged; then multiply by the derivative of the inner function.

Leibniz notation. If $u = g(x)$ and $F = f(u)$, then

$$\frac{dF}{dx} = \frac{df}{du}\frac{du}{dx} = \frac{df}{dg}\frac{dg}{dx}.$$

Here the outer derivative is evaluated at $u = g(x)$. Trigonometric derivatives use radian measure.

Color key: Factors are colored from the outside inward: first layer, second, third, fourth. Combined final results are left in black.

Using the chain rule with a table

Example 1. Use the following values to compute the derivatives below.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
0	2	3	1	-2
1	4	5	0	6
2	1	-1	3	4
3	2	-2	1	5

(i) Find $(f \circ g)'(0)$.

Evaluate the inner function first. The table gives $g(0) = 1$, so

$$\begin{aligned}(f \circ g)'(0) &= f'(g(0))\, g'(0) \\ &= f'(1)\,(-2) \\ &= 5\,(-2) = -10.\end{aligned}$$

(ii) Find $(g \circ f)'(0)$.

Reverse the roles. Now f is the inner function. Since $f(0) = 2$,

$$\begin{aligned}(g \circ f)'(0) &= g'(f(0))\, f'(0) \\ &= g'(2)\,(3) \\ &= 4\,(3) = 12.\end{aligned}$$

Practice

(iii) Find $(f \circ g)'(2)$.

$$\begin{aligned}(f \circ g)'(2) &= f'(g(2))\, g'(2) \\ &= f'(3)\,(4) = (-2)\,(4) = -8.\end{aligned}$$

(iv) Find $(g \circ f)'(2)$.

$$\begin{aligned}(g \circ f)'(2) &= g'(f(2))\, f'(2) \\ &= g'(1)\,(-1) = 6\,(-1) = -6.\end{aligned}$$

The outer derivative is evaluated at the *output of the inner function*.

Powers of functions

Question. How do we differentiate $(g(x))^n$?

Example 0. Recall the power rule:

$$\frac{d}{dx}(x^4) = 4x^3.$$

Example 1. Differentiate $(x+10)^4$.

$$\begin{aligned}\frac{d}{dx}(x+10)^4 &= 4(x+10)^3\,\frac{d}{dx}(x+10)\\ &= 4(x+10)^3\cdot 1 = 4(x+10)^3.\end{aligned}$$

Here $f(u) = u^4$ and $g(x) = x+10$.

Example 2. Differentiate $\cos^4 x$.

$$\frac{d}{dx}(\cos x)^4 = 4\cos^3 x\,(-\sin x) = -4\cos^3 x \sin x.$$

Example 3. Differentiate $\sin^2 x$.

$$\frac{d}{dx}(\sin x)^2 = 2\sin x\cos x.$$

Example 4. Differentiate $\sqrt{1+x^2}$.

$$\begin{aligned}\frac{d}{dx}(1+x^2)^{1/2} &= \frac{1}{2}(1+x^2)^{-1/2}\,(2x)\\ &= \frac{x}{\sqrt{1+x^2}}.\end{aligned}$$

Corollary (Power rule for a function)**.** *If g is differentiable, then*

$$\frac{d}{dx}(g(x))^n = n(g(x))^{n-1}\,g'(x),$$

wherever the power function is differentiable at $g(x)$.

Proof. Apply the chain rule with outer function $f(u) = u^n$. □

Practice. Suppose $y = (g(x))^2$, $g'(2) = 3$, and $y'(2) = 24$. Find $g(2)$.

Use the chain rule.

$$y' = 2g(x)\,g'(x), \qquad 24 = 2g(2)\cdot 3 = 6g(2).$$

Therefore $g(2) = 4$.

Combining differentiation rules

Example 1. Differentiate $y = x^3 \cos^2 x$.

Product rule, then chain rule.

$$y' = 3x^2 \cos^2 x + x^3 \frac{d}{dx}(\cos^2 x).$$

For the second factor,

$$\begin{aligned}\frac{d}{dx}(\cos^2 x) &= 2\cos x \, \frac{d}{dx}(\cos x) \\ &= 2\cos x \, (-\sin x) = -2\sin x \cos x.\end{aligned}$$

Substituting gives

$$\begin{aligned}y' &= 3x^2 \cos^2 x + x^3 [2\cos x \, (-\sin x)] \\ &= 3x^2 \cos^2 x - 2x^3 \sin x \cos x.\end{aligned}$$

Example 2. Differentiate $y = \left(\dfrac{x+1}{2x-1}\right)^2$.

Chain rule, then quotient rule. The outer function squares its input, so

$$y' = 2\left(\frac{x+1}{2x-1}\right)\frac{d}{dx}\left(\frac{x+1}{2x-1}\right).$$

Differentiate the inner quotient:

$$\begin{aligned}\frac{d}{dx}\left(\frac{x+1}{2x-1}\right) &= \frac{(2x-1)(1) - (x+1)(2)}{(2x-1)^2} \\ &= \frac{2x - 1 - 2x - 2}{(2x-1)^2} \\ &= \frac{-3}{(2x-1)^2}.\end{aligned}$$

Therefore

$$\begin{aligned}y' &= 2\left(\frac{x+1}{2x-1}\right)\left(\frac{-3}{(2x-1)^2}\right) \\ &= -\frac{6(x+1)}{(2x-1)^3}, \qquad x \neq \frac{1}{2}.\end{aligned}$$

Three or more layers

Question. What happens when there are three or more layers? Apply the chain rule repeatedly, working from the outside inward.

1. Differentiate $y = \sin^3(\cos(2x))$.

Read the expression as $[\sin(\cos(2x))]^3$. From outside inward: cube, sine, cosine, $2x$. Its layers are

$$x \longmapsto 2x \longmapsto \cos(2x) \longmapsto \sin(\cos(2x)) \longmapsto [\sin(\cos(2x))]^3.$$

Differentiate one layer at a time.

$$\begin{aligned} y' &= 3\sin^2(\cos(2x)) \frac{d}{dx}[\sin(\cos(2x))] \\ &= 3\sin^2(\cos(2x)) \cos(\cos(2x)) \frac{d}{dx}[\cos(2x)] \\ &= 3\sin^2(\cos(2x)) \cos(\cos(2x)) (-\sin(2x)) \frac{d}{dx}(2x) \\ &= 3\sin^2(\cos(2x)) \cos(\cos(2x)) (-\sin(2x)) \, 2 \\ &= -6\sin^2(\cos(2x)) \cos(\cos(2x)) \sin(2x). \end{aligned}$$

Each layer contributes one factor, including the innermost derivative $\frac{d}{dx}(2x) = 2$.

2. Differentiate $y = \sin(\cos(\tan x))$.

From outside inward: sine, cosine, tangent. The layers are

$$x \longmapsto \tan x \longmapsto \cos(\tan x) \longmapsto \sin(\cos(\tan x)).$$

Differentiate one layer at a time.

$$\begin{aligned} y' &= \cos(\cos(\tan x)) \frac{d}{dx}[\cos(\tan x)] \\ &= \cos(\cos(\tan x)) (-\sin(\tan x)) \frac{d}{dx}(\tan x) \\ &= \cos(\cos(\tan x)) (-\sin(\tan x)) \sec^2 x \\ &= -\cos(\cos(\tan x)) \sin(\tan x) \sec^2 x. \end{aligned}$$

The formula applies wherever $\tan x$ is defined.

Practice

1. Differentiate $y = x^2 \sin^3(2x)$.

Product rule and chain rule.

$$\begin{aligned} y' &= 2x \sin^3(2x) + x^2 \frac{d}{dx}\left[\sin^3(2x)\right] \\ &= 2x \sin^3(2x) + x^2 \left[3 \sin^2(2x) \cos(2x) \cdot 2\right] \\ &= 2x \sin^3(2x) + 6x^2 \sin^2(2x) \cos(2x). \end{aligned}$$

2. Differentiate $y = \dfrac{1}{\tan^2(2x) + 1}$.

Write the reciprocal as a power.

$$y = \left[\tan^2(2x) + 1\right]^{-1}.$$

Then

$$\begin{aligned} y' &= -\left[\tan^2(2x) + 1\right]^{-2} \frac{d}{dx}\left[\tan^2(2x) + 1\right] \\ &= -\left[\tan^2(2x) + 1\right]^{-2} \left[2 \tan(2x) \sec^2(2x) \cdot 2\right] \\ &= -\frac{4 \tan(2x) \sec^2(2x)}{\left[\tan^2(2x) + 1\right]^2}. \end{aligned}$$

This formula is valid where the original function is defined, namely where $\cos(2x) \neq 0$.

3. Find the tangent line to $f(x) = \sqrt{1 + 3x^2}$ at $x = 1$.

Find the point and the slope. First, $f(1) = \sqrt{4} = 2$. By the chain rule,

$$\begin{aligned} f'(x) &= \frac{1}{2}(1 + 3x^2)^{-1/2} \, (6x) \\ &= \frac{3x}{\sqrt{1 + 3x^2}}. \end{aligned}$$

Thus $f'(1) = \frac{3}{2}$. The tangent line is

$$y - 2 = \frac{3}{2}(x - 1), \qquad \text{or} \qquad y = \frac{3}{2}x + \frac{1}{2}.$$